



FIGURE 5.3.18 LRC series circuit

#### 5.3.4 SERIES CIRCUIT ANALOGUE

**■ LRC Series Circuits** As was mentioned in the introduction to this chapter, many different physical systems can be described by a linear second-order differential equation similar to the differential equation of forced motion with damping:

$$m \frac{d^2 x}{dt^2} + b \frac{dx}{dt} + kx = F(t). \quad (52)$$

If  $i(t)$  denotes current in the LRC series electrical circuit shown in Figure 5.3.18, then the voltage drops across the inductor, resistor, and capacitor are as shown in Figure 5.3.4. By Kirchhoff's second law, the sum of these voltages equals the voltage  $E(t)$  impressed on the circuit, that is,

$$L \frac{di}{dt} + Ri + \frac{1}{C} i = E(t). \quad (53)$$

But the charge  $q(t)$  on the capacitor is related to the current  $i(t)$  by  $i = dq/dt$ , so (53) becomes the linear second-order differential equation

$$L \frac{d^2 q}{dt^2} + R \frac{dq}{dt} + \frac{1}{C} q = E(t). \quad (54)$$

The correspondence used in the analysis of circuits is similar to that used to describe spring-mass systems.

If  $E(t) = E$ , the electrical analogue of the circuit can be said to be *free*. Because the secondary equation for (54) is  $Lm^2 + Rm + 1/C = 0$ , there will be three kinds of behavior with  $R \neq 0$ , depending on the value of the discriminant  $R^2 - 4L/C$ . We say that the circuit is

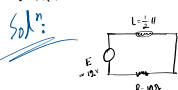
**overdamped** if  $R^2 - 4L/C > 0$ ,  
**critically damped** if  $R^2 - 4L/C = 0$ ,  
and **underdamped** if  $R^2 - 4L/C < 0$ .

In each of these three cases the general solution of (54) contains the factor  $e^{-Rt/2L}$ , so  $q(t) \rightarrow 0$  as  $t \rightarrow \infty$ . In the underdamped case when  $q(t) \neq 0$ , the charge on the capacitor oscillates as it decays; in other words, the capacitor is charging and discharging as  $t \rightarrow \infty$ . When  $R(t) = 0$  and  $R = 0$ , the circuit is said to be *undamped*, and the electrical vibrations do not approach zero as  $t$  increases without bound; the response of the circuit is *simple harmonic*.

Expt 7 A 12-volt battery is connected to a series circuit in which

the inductance is  $\frac{1}{2}$  henrys and the resistance is 20 ohms.

Determine the current  $i(t)$  if  $i(0) = 0$ .



$$\frac{1}{2} \frac{di}{dt} + 20i = 12, \quad i(0) = 0$$

General form

$$\Rightarrow \frac{di}{dt} + 20i = 24, \quad i(0) = 0$$

Standard form

$$\frac{di}{dt} + P(t)i = Q(t)$$

$$\Rightarrow IF = e^{\int P(t) dt}$$

$$\Rightarrow IF = e^{\int 20 dt} = e^{20t}$$

$$\text{Now, } e^{20t} \left[ \frac{di}{dt} + 20i \right] = 24e^{20t}$$

$$\Rightarrow e^{20t} \frac{di}{dt} + 20i \cdot e^{20t} = 24e^{20t}$$

$$\Rightarrow [e^{20t} \cdot i] dt = 24 \int e^{20t} dt$$

$$\Rightarrow e^{20t} \cdot i = 24 \frac{e^{20t}}{20} + c$$

$$\Rightarrow i(t) = \frac{6}{5} + ce^{-20t}$$

$$\Rightarrow i(1) = \frac{6}{5} (1 - e^{-20}) \quad \left\{ \begin{array}{l} \text{if } i(0) = 0 \\ \Rightarrow c = -\frac{6}{5} \end{array} \right.$$

#### EXAMPLE 9 Underdamped Series Circuit

Find the charge  $q(t)$  on the capacitor in an LRC-series circuit when  $L = 0.25$  henry (h),  $R = 10$  ohms ( $\Omega$ ),  $C = 0.001$  farad (f),  $E(t) = 0$ ,  $q(0) = q_0$  coulombs (C), and  $i(0) = 0$ .

Sol<sup>n</sup>:

$$\left. \begin{array}{l} \text{Here, } L = 0.25 \text{ H} \\ R = 10 \Omega \\ C = 0.001 \text{ F} \end{array} \right\} \begin{array}{l} E(t) = 0 \\ q(0) = q_0 \\ i(0) = 0 \end{array}$$

$$\text{We know, } L \frac{di}{dt} + Ri + \frac{1}{C} q = E(t)$$

$$\Rightarrow L \frac{d^2 q}{dt^2} + R \frac{dq}{dt} + \frac{1}{C} q = E(t)$$

$$\Rightarrow L q''(t) + R q'(t) + \frac{1}{C} q(t) = E(t)$$

$$\Rightarrow 0.25 q''(t) + 10 q'(t) + 1000 q(t) = 0$$



$$\Rightarrow 0.25 q''(t) + 10 q'(t) + 1000 q(t) = 0$$

$$\Rightarrow q''(t) + 40 q'(t) + 4000 q(t) = 0$$

A homogeneous  
2<sup>nd</sup> Order linear DE

$$\rightarrow m^2 + 40m + 4000 = 0$$

$$\Rightarrow m = -20 \pm 60i$$

$$\therefore q(t) = e^{-20t} [c_1 \cos(60t) + c_2 \sin(60t)]$$

Rules to find a complementary function

$$(a_n D^n + a_{n-1} D^{n-1} + a_{n-2} D^{n-2} + \dots + a_0) y = 0 \rightarrow \textcircled{1}$$

Case 1  $\rightarrow$  If roots are real & distinct ( $m_1 \neq m_2$ )  
 $CF = c_1 e^{m_1 x} + c_2 e^{m_2 x} + \dots + c_n e^{m_n x}$

Case 2  $\rightarrow$  When roots are real & equal ( $m_1 = m_2 = m$ )  
 $CF = (c_1 + c_2 x) e^{mx} + c_3 e^{m_3 x} + \dots + c_n e^{m_n x}$

Case 3  $\rightarrow$  When roots are imaginary  
 $m_1 = \alpha + i\beta, m_2 = \alpha - i\beta$   
 $CF = e^{\alpha x} (c_1 \cos \beta x + c_2 \sin \beta x)$

Now,  $q(0) = q_0$

$$\therefore q_0 = e^0 [c_1 \cos 0 + 0]$$

$$\therefore c_1 = q_0$$

$$\therefore q(t) = e^{-20t} [q_0 \cos(60t) + c_2 \sin(60t)]$$

$$\hookrightarrow i(t) = q'(t) = e^{-20t} [-60 q_0 \sin(60t) + 60 c_2 \cos(60t)]$$

$$- 20 e^{-20t} [q_0 \cos 60t + c_2 \sin 60t]$$

$$\Rightarrow i(0) = 0 \Rightarrow 0 = 60 c_2 - 20 q_0$$



$$\Rightarrow C_2 = \frac{1}{3} q_0$$

$$\therefore q(t) = e^{-20t} \left[ q_0 \cos(60t) + \frac{q_0}{3} \sin(60t) \right]$$

### EXAMPLE 10 Steady-State Current

Find the steady-state solution  $q_p(t)$  and the **steady-state current** in an *LRC*-series circuit when the impressed voltage is  $E(t) = E_0 \sin \gamma t$ .

Sol<sup>n</sup>:

$$L \frac{d^2 q}{dt^2} + R \frac{dq}{dt} + \frac{1}{C} q = E_0 \sin \gamma t$$

Using the method of undetermined coefficient, the particular solution in the

$$q_p(t) = A \sin \gamma t + B \cos \gamma t$$

From the method of undetermined coefficients

$$A = \frac{E_0 \left( L\gamma - \frac{1}{C\gamma} \right)}{-\gamma \left( L^2 \gamma^2 - \frac{2L}{C} + \frac{1}{C^2 \gamma^2} + R^2 \right)}, \quad B = \frac{E_0 R}{-\gamma \left( L^2 \gamma^2 - \frac{2L}{C} + \frac{1}{C^2 \gamma^2} + R^2 \right)}$$

Memorize them.  
It's a hassle to show the proof

$$\text{Assuming } X = L\gamma - \frac{1}{C\gamma} \Rightarrow X^2 = L^2 \gamma^2 - \frac{2L}{C} + \frac{1}{C^2 \gamma^2} + R^2$$

0.01 m

0.01 m

$$v = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} \Rightarrow v = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$z = \sqrt{x^2 + R^2} \Rightarrow z^2 = x^2 + R^2$$

$$\therefore A = \frac{F_0 x}{-r z^2} ; B = \frac{F_0 R}{-r z^2}$$

$$\therefore q(t) = -\frac{F_0 x}{r z^2} \sin rt - \frac{F_0 R}{r z^2} \cos rt$$

$$\Rightarrow i(t) = q'(t) = \frac{F_0}{z} \left( \frac{R}{z} \sin rt - \frac{x}{z} \cos rt \right)$$

[Baad en math]

